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## **LAB 3**

# **Municipal Solid Waste Route Optimizer**

### **Architecture Selection**

**We chose Layered Architecture for the Municipal Solid Waste Route Optimizer.**

### **Architectural Choice**

Layered Architecture was selected because the system can be divided into Presentation, Business Logic and Data layers, providing clear separation of responsibilities.

### **Reason 1**

The architecture separates the user interface from route optimization and collection management, making the system easier to maintain and modify.

### **Reason 2**

The data layer centralizes bin, route, truck and collection information, allowing the business components to access data in an organized manner.

### **Security Advantage**

The layered structure prevents the user interface from directly accessing the database, allowing access control to be implemented between the application and data layers.

### **Performance Benefit**

The Route Optimization component can process bin fill-level information and calculate routes independently of the user interface, allowing route generation to be performed efficiently.